Exam Date & Time: 04-Jan-2025 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

Biomaterial-characterization techniques [BME 4052]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions. Section Duration: 180 mins

Answer all questions

1A)	Compare the different mechanical properties of materials and predict the change in properties with the inclusion of crystal defects.	(4)
1B)	Discuss the various conjugate focal planes in optical microscope.	(2)
1C)	Explain the different configurations in X-ray diffraction (XRD), including their principles, advantages, and specific applications in material analysis.	(4)
2A)	A researcher is analyzing the X-ray diffraction (XRD) pattern of a nanomaterial and observe a broadening of the diffraction peaks compared to bulk material. Analyze the possible causes of peak broadening in the XRD pattern.	(3)
2B)	A researcher is given two protein solutions, A and B, with known molar absorptivity's at 280 nm. Solution A has an absorbance of 0.85 and solution B has an absorbance of 1.10, both measured using a 1 cm path length cuvette. Analyze the data to determine which solution has a higher protein concentration and explain the relationship between absorbance, molar absorptivity, and concentration. How would changes in path length or the protein's molar absorptivity affect the concentration calculations for these solutions?	(4)
2C)	A researcher needs to confirm and characterize the fluorescence of a new nanomaterial for imaging applications. Outline how you would use fluorescence spectroscopy to study this nanomaterial. Briefly describe sample preparation, choosing an excitation wavelength, and interpreting the emission spectrum.	(3)
3A)	Describe how the contact angle of a biomaterial changes with respect to surface roughness and contamination.	(3)
3B)	Describe the working principle of Dynamic Light Scattering (DLS) and outline its key components.	(3)
3C)	Analyze the factors that influence the zeta potential of colloidal suspensions.	(4)
4A)	Explain the principle and components used in column chromatography.	(3)
4B)	Distinguish between isocratic and gradient elution in liquid chromatography.	(3)

4C)	Elaborate the various ionization methods used in Mass spectroscopy.	(4)
5A)	Analyze the rationale behind using an electron gun in Time-of-Flight Secondary Ion Mass Spectrometry (TOF-SIMS) and explain its contribution to the overall analysis.	(3)
5B)	Differentiate between Differential Thermal Analysis (DTA) and Differential Scanning Calorimetry (DSC).	(4)
5C)	A scientist characterized a surface modified with amine functional group using X-ray photoelectron spectroscopy (XPS). He characterized the surface and found apart from amine groups other functional groups like OH were present. Examine the possible reason why OH groups were characterized. Propose method by which the scientist can more accurately characterize the surface.	(3)

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